# RAMCO INSTITUTE OF TECHNOLOGY

## RAJAPALAYAM

**Department of Artificial Intelligence and Data Science**

**Academic Year: 2025- 2026 (Odd Semester)**

**Active Learning Best practices: JIGSAW-Cooperative Learning Technique**

**Degree, Semester & Branch: VII Sem. B.Tech. Artificial Intelligence and Data Science**

**Course Code & Title: - GE3755 & Knowledge Management**

**Name of the Faculty member: Ms.V. Logapriya/Assistant Professor-AI & DS**

---

**Theme of discussion:**

**Topics Covered: Unit IV**

**Date and Time:06.10.2025 & 9.45 A.M to 10.30 A.M**

---

**Course Outcome: CO4**

- **Topic Learning Outcome: TLO16**
- **Activity Chosen: JIGSAW**

---

**Active Learning Best practices:** JIGSAW

**Topic: Knowledge Management in the Health Sciences.**

## Learning Outcomes

- **Critical Thinking:**
  Students will critically analyze how knowledge management processes—such as knowledge capture, storage, sharing, and application—enhance decision-making, patient safety, and innovation in healthcare organizations.
- **Communication Skills:**
  Students will demonstrate structured oral communication through peer teaching, group presentations, and explanation of healthcare KM systems in simple, organized terms.
- **Collaboration and Teamwork:**
  Students will practice cooperative learning through the JIGSAW method by becoming "experts" in subtopics and sharing their knowledge with peers to build a collective understanding.
- **Knowledge Application:**
  Students will relate KM theories to real-world healthcare systems by identifying how digital tools, AI, and data analytics contribute to effective knowledge sharing in the health sciences.

## Procedure

🎬 Faculty distributes pre-recorded lectures and reading materials focusing on:

- Core KM concepts in healthcare
- Digital tools for KM (e.g., electronic health records, AI, data analytics)
- Real-world applications and challenges of KM in healthcare settings

🎬 Students prepare by:

- Summarizing key points
- Reflecting on examples where KM impacted healthcare outcomes
- Preparing to become "experts" on their assigned subtopic

## Step 1: Formation of Expert Groups (10 minutes)

- Students are divided into expert groups, each assigned a KM subtopic:
  - Group 1: Knowledge Capture and Storage in Health Sciences
  - Group 2: Knowledge Sharing and Communication Tools (e.g., Listserv, meetings)
  - Group 3: Application of KM in Clinical Decision Making and Patient Safety
  - Group 4: Role of Digital Tools, AI, and Data Analytics in Healthcare KM
  - Group 5: Challenges and privacy measures in Healthcare KM
- Each expert group studies and discusses their topic in depth, preparing to teach peers.

## Step 2: Jigsaw Teaching Phase (10 minutes)

- Students reorganize into mixed groups, with one member from each expert group.
- Each student teaches their subtopic to their new group members, facilitating peer learning.
- Groups collaboratively build a comprehensive understanding of KM in health sciences.

## Step 3: Group Discussion & Synthesis (15 minutes)

- Groups discuss how KM processes interrelate and contribute to healthcare quality improvement.
- Develop a visual summary/chart illustrating:
  - How KM enhances healthcare outcomes

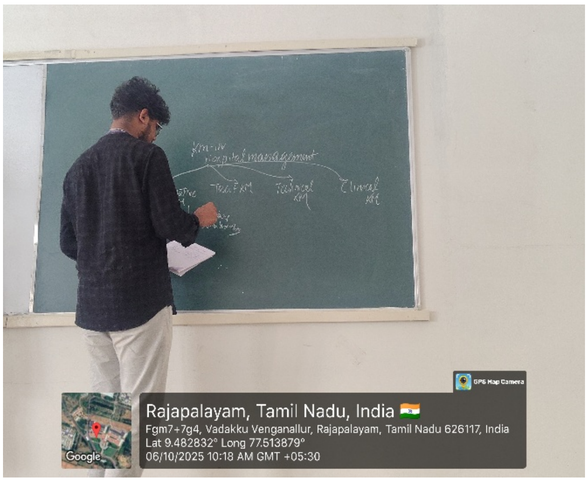
- Challenges and best practices in healthcare KM
- Impact of technology and teamwork on KM success

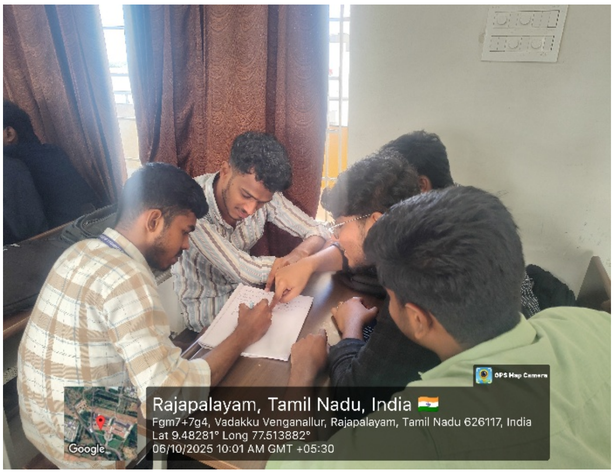
## Step 4: Reflection & Wrap-Up (10 minutes)

- Students individually reflect on:
  - How they can apply KM concepts in future healthcare roles
  - Which KM tools or processes they see as most impactful
- Faculty summarizes key insights and reinforces the importance of KM in advancing health sciences.

## Glimpses





## Challenges & Strategies

- **Challenge:** Varying prior knowledge of KM tools and health sciences concepts.
  **Strategy:** Use clear pre-class materials and guided expert group discussions.
- **Challenge:** Ensuring equal participation.
  **Strategy:** Assign roles within groups (e.g., presenter, note-taker, facilitator).
- **Challenge:** Time management for thorough coverage.
  **Strategy:** Structured timing and clear instructions.

## Benefits

- Builds higher-order thinking by encouraging analysis, synthesis, and evaluation.
- Enhances communication and teaching skills.
- Simulates real-world interdisciplinary teamwork in healthcare.
- Demonstrates practical value of KM in improving patient care and organizational efficiency.

## Observations

- **Enhanced Understanding of KM Concepts:**
  Students demonstrated a solid grasp of core knowledge management processes—capture, storage, sharing, and application—specifically how these apply to healthcare settings. The expert groups showed strong command of their subtopics, reflecting effective pre-class preparation.
- **Effective Peer Teaching:**
  The JIGSAW method facilitated clear, organized peer teaching. Students explained complex KM tools and theories in accessible terms, helping their group members build an integrated understanding of the topic.
- **Active Engagement and Participation:**
  Role-play activities, especially simulations of Listserv posts and meetings, significantly increased student involvement and motivation. The interactive nature helped students experience real-world communication challenges and advantages firsthand.
- **Critical Thinking and Analytical Skills:**
  Students effectively compared asynchronous and synchronous knowledge sharing methods, discussing practical implications like participation styles, communication permanence, and accessibility in healthcare organizations.
- **Collaboration and Teamwork:**
  Students worked cooperatively within their expert and mixed groups, sharing responsibility and encouraging balanced participation. Assigning roles such as leader and note-taker ensured structured teamwork.
- **Application of Technology in KM:**
  Discussions revealed increasing student awareness of how digital tools and AI integrate into healthcare KM, demonstrating relevance to current and future professional environments.

## Student Response (Expected)

- **High Engagement and Enthusiasm:**
  Most students reported enjoying the interactive format, especially the opportunity to teach peers and engage in role-play exercises. They appreciated learning from classmates' expertise.
- **Improved Confidence:**
  Students felt more confident explaining KM concepts and the practical applications of digital tools in healthcare after teaching their subtopics.
- **Deeper Understanding of Communication Dynamics:**
  Through comparison of asynchronous and synchronous communication, students gained clearer insight into how different platforms influence knowledge sharing effectiveness.
- **Recognition of Real-World Relevance:**
  Students valued connecting theoretical KM models to actual healthcare settings, noting the importance of knowledge sharing in improving patient care and safety.
- **Appreciation of Collaborative Learning:**
  Many students highlighted how the JIGSAW method encouraged them to listen actively and respect diverse perspectives, fostering a supportive learning environment.
- **Constructive Feedback:**
  Some students suggested allocating slightly more time for group discussions and practice presentations to deepen understanding and improve delivery.

# RAMCO INSTITUTE OF TECHNOLOGY

## RAJAPALAYAM

**Department of Artificial Intelligence and Data Science**

**Academic Year: 2025- 2026 (Odd Semester)**

**FEEDBACK**

**Active Learning Best practices: JIGSAW-Cooperative Learning Technique**

**Degree, Semester & Branch: VII Sem. B.Tech. Artificial Intelligence and Data Science**

**Course Code & Title: - GE3755 & Knowledge Management**

**Name of the Faculty member: Ms.V. Logapriya/Assistant Professor-AI & DS**

---

**Theme of discussion:**

**Topics Covered: Unit IV**

**Date and Time:06.10.2025 & 9.45 A.M to 10.30 A.M**

---

**Feedback collected in class and also through online**

**Gform Link** : https://forms.gle/qisRc32N26KzTEkZA

---

**Feedback Questions:**

**1.** Did the active learning method used in the session engage your interest in the topic? **Yes** **No**

**2.** How did the active learning method enhance your understanding of the topic?

**Excellent** **Good** **Satisfactory**

**3**. Did the active learning method encourage active participation and communication?

**Yes** **No**

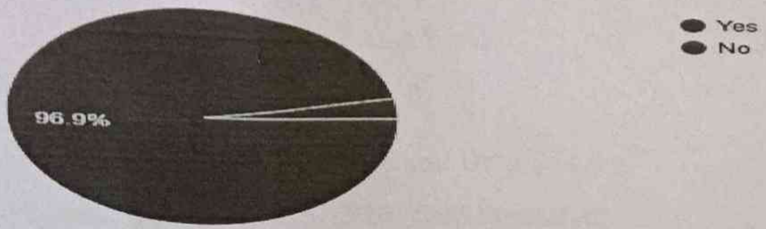
4. Did the active learning method prompt you to think more deeply or critically about the topic?

**Yes** **No**

## Feedback Analysis:

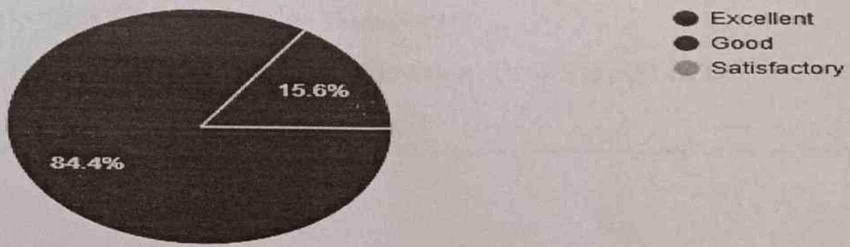
1.Did the active learning method used in the session engage your interest in the topic?

32 responses



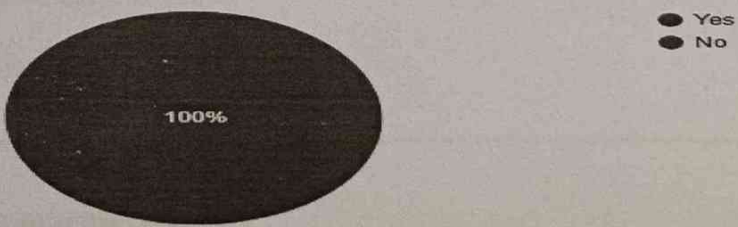

2.How did the active learning method enhance your understanding of the topic?

32 responses



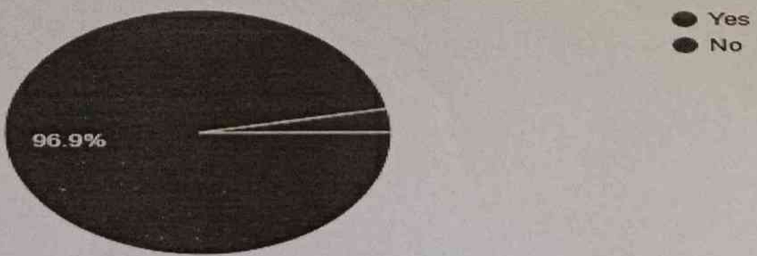

3.Did the active learning method encourage active participation and communication?

32 responses




4. Did the active learning method prompt you to think more deeply or critically about the topic?

32 responses




8/10/25

**FACULTY INCHARGE**

8/10/25

**HOD/AD**